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Matrix Problem

In [1]: import numpy as np

In [2]: A=np.array([1,2,3])

In [3]: print(A)

[1 2 3]

In [4]: A = np.array([[1, 2], [3, 4]])
        B = np.array([[2, 0], [1, 2]])

In [5]: print(A)

[[1 2]
 [3 4]]

In [6]: print(B)

[[2 0]
 [1 2]]

In [8]: print(A.shape)

(2, 2)

In [9]: print(B.shape)

(2, 2)

In [11]: C=A+B

In [12]: print(C)

[[3 2]
 [4 6]]

In [13]: D=np.dot(A,B)

In [14]: print(D)

[[ 4  4]
 [10  8]]

Matrics Sample

In [15]: import numpy as np

In [17]: matrix1=np.array([[1,3],[5,7]])

In [19]: matrix2=np.array([[2,3,5],[7,14,21],[1,3,5]])

In [20]: print(matrix1)
        print(matrix2)

[[1 3]
 [5 7]]
[[ 2  3  5]
 [ 7 14 21]
 [ 1  3  5]]

In [21]: print(matrix1.shape)
        print(matrix2.shape)

(2, 2)
(3, 3)

In [28]: matrix3=np.array([[2,6],[6,8]])

In [29]: print(matrix3)

[[2 6]
 [6 8]]

In [31]: product=np.dot(matrix1, matrix3)
        print(product)

[[20 30]
 [52 86]]

In [33]: Transpose=np.transpose(matrix2)
        print(Transpose)

[[ 2  7  1]
 [ 3 14  3]
 [ 5 21  5]]

In [ ]:

Random Number Generator

In [34]: import numpy as np

In [35]: rand_num=np.random.rand()

In [36]: print(rand_num)

0.22834664261660398

In [37]: rand_matrix=np.random.rand(3,3)

In [38]: print(rand_matrix)

[[0.36586206 0.0856131 0.73093193]
 [0.669627   0.28189071 0.42493097]
 [0.93951073 0.13517642 0.35825047]]

In [39]: rand_int_matrix=np.random.randint(0,10, size=(4,4))
        print(rand_int_matrix)

[[5 2 0 2]
 [1 3 0 9]
 [1 5 7 8]
 [0 4 2 3]]

In [40]: rand_normal=np.random.randn(2,5)
        print(rand_normal)

[[-0.34761519 -0.42054212 -0.63307299 0.33608174 -1.36272074]
 [-1.16035273 0.03476731 -2.63286348 1.37189693 0.86285384]]

In [ ]:

Arithmetic Operations

In [42]: mx1 = np.array([[5, 10], [15, 20]])
        mx2 = np.array([[25, 30], [35, 40]])

In [43]: print(mx1)
        print(mx2)

[[ 5 10]
 [15 20]]
[[25 30]
 [35 40]]

In [45]: print("Addition of Two matrices:")
        print(np.add(mx1,mx2))

Addition of Two matrices:
[[30 40]
 [50 60]]

In [48]: print("Substraction of Two Matrices:")
        print(np.subtract(mx1,mx2))

Substraction of Two Matrices:
[[-20 -20]
 [-20 -20]]

In [ ]:

Aggragation Function

In [49]: import numpy as np

In [50]: mx = np.array([[5, 10], [15, 20]])

        print("Matrix =\n",mx)

Matrix =
[[ 5 10]
 [15 20]]

In [51]: print("Summation of elements:")
        print(np.sum(mx))

Summation of elements:
50

In [52]: print("Column wise summation:")
        print(np.sum(mx,axis=0))

Column wise summation:
[20 30]

In [54]: print("Row wise summation:")
        print(np.sum(mx,axis=1))

Row wise summation:
[15 35]

In [ ]:

Matrix Comparisons

In [55]: import numpy as np

In [56]: arr1 = np.array([1, 5, 3, 7])
        arr2 = np.array([2, 5, 1, 8])
        print(arr1)
        print(arr2)

[1 5 3 7]
[2 5 1 8]

In [57]: result=arr1>arr2
        print(result)

[False False  True False]

In [58]: count=np.sum(arr1>4)
        print(count)

2

In [59]: arr1 = np.array([1, 5, 3, 7])
        arr2 = np.array([1, 2, 3, 8])

        print(arr1)
        print(arr2)

[1 5 3 7]
[1 2 3 8]

In [61]: indices=np.where(arr1==arr2)[0]
        print(indices)

[0 2]

In [62]: arr = np.array([1, 3, 5, -2])

        all_positive = np.all(arr > 0)
        print(all_positive)

False

In [63]: arr = np.array([2, 8, 1, 5, 3])
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filtered = arr[arr < 5]
print(filtered)
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[2 1 3]
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In []: